



Economic Assessment of an Off-Grid Solar PV

PREPARED FOR THE UNHCR

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1. Executive Summary

This report presents a techno-economic assessment conducted by *Energy Solutions Consulting Group* for the United Nations High Commissioner for Refugees (UNHCR). The study evaluates the feasibility of developing an **off-grid solar photovoltaic (PV) and battery energy storage system** to supply reliable and sustainable electricity to a refugee camp housing approximately **100,000 people near Damascus, Syria**. The objective is to provide a clean and independent alternative to conventional diesel generator systems, which are costly, polluting, and logistically challenging to maintain.

A detailed **load analysis** was performed for residential units, clinics, schools, street lighting, and water pumping systems. Based on these estimates, the camp's **average daily energy demand** was determined, and the solar PV system and battery storage were sized accordingly. The analysis assumed a **solar irradiance of 5.5 kWh/m²/day**, **PV panel cost of \$300/kWp**, **battery cost of \$200/kWh** (10-year lifetime), and **diesel fuel cost of \$1.20 per liter**, referencing data from IRENA and regional energy studies.

The optimized configuration consists of an estimated **30 MWp solar PV system** paired with approximately **100 MWh of lithium-ion battery storage**, capable of supplying continuous power to all camp facilities. The total capital cost is projected at **\$9.5 million for PV modules** and **\$20 million for energy storage**, with additional costs for inverters, wiring, and installation bringing the total investment to roughly **\$35–40 million**. Over a **20-year project lifetime**, the solar-battery system demonstrates a **38% lower total cost of ownership** compared to diesel generation, with a **payback period of approximately 8 years**.

Beyond economic advantages, the proposed system eliminates an estimated **50,000 tons of CO₂ emissions annually**, reduces noise and fuel logistics, and enhances long-term energy security for the camp.

It is therefore recommended that the UNHCR proceed with a **phased implementation** of the solar PV and battery energy storage system, starting with a **10**

MW pilot installation to validate design assumptions, followed by full deployment. The project represents a technically feasible and economically favorable solution that aligns with UNHCR's sustainability and humanitarian goals.

2. Introduction

The Syrian refugee crisis has resulted in millions of displaced individuals, many of whom are returning to areas near Damascus. Providing reliable, sustainable, and cost-effective electricity to refugee settlements is a major humanitarian challenge. Traditional diesel generators are widely used in these settings but are associated with high operational costs, fuel supply vulnerabilities, noise pollution, and significant greenhouse gas emissions.

In response to this challenge, the United Nations High Commissioner for Refugees (UNHCR) commissioned *Energy Solutions Consulting Group* to evaluate the feasibility of deploying an **off-grid solar photovoltaic (PV) system combined with battery energy storage** to supply power to a refugee camp accommodating approximately **100,000 people**. This study aims to support decision-making by providing a detailed assessment of both technical and economic aspects, including system sizing, cost analysis, and comparison with conventional diesel generation.

The report focuses on:

- Assessing the **technical feasibility** of meeting the camp's electricity demand using solar PV and energy storage.
- Conducting a **economic evaluation** to determine cost-effectiveness relative to diesel generators, considering capital expenditures, operations and maintenance, fuel costs, and system lifetime.
- Providing **recommendations** for implementation, including system design, sizing, and potential phased deployment strategies.

By evaluating renewable energy alternatives in a refugee settlement context, this study aims to provide a solution that enhances energy security, reduces environmental impact, and supports sustainable humanitarian operations.

3. Objective and Scope

3.1 Objective

The primary objective of this report is to provide a **techno-economic assessment** of an off-grid solar photovoltaic (PV) system coupled with battery energy storage to serve a refugee camp near Damascus, Syria, accommodating approximately **100,000 people**. The study evaluates both the **technical feasibility** of supplying reliable electricity and the **economic viability** of the system relative to conventional diesel generators.

Specific Objectives

1. **Load Assessment:** Quantify electricity demand for all camp facilities, including residential prefab units, clinics, schools, water pumping stations, and street lighting.
2. **System Design:** Determine optimal sizing for solar PV arrays, inverters, and battery storage to meet daily and peak energy requirements.
3. **Economic Evaluation:** Compare the total cost of ownership for the solar PV + battery system versus diesel generation, including capital expenditure, operations and maintenance, fuel costs, and projected system lifetime.
4. **Environmental Assessment:** Evaluate potential reductions in greenhouse gas emissions and other environmental benefits associated with solar PV adoption.
5. **Recommendations:** Provide actionable guidance on implementation strategies, phased deployment, and further study requirements.

3.2 Scope

Included	Excluded
Residential energy use (prefab units)	Grid integration or connection to existing utility infrastructure

Clinics, schools, street lighting, and water pumping loads	Detailed civil construction or site-specific engineering design
Off-grid solar PV system sizing	Security, distribution, and metering beyond the system boundaries
Battery energy storage sizing and configuration	
Cost and LCOE analysis against diesel generators	

This scope ensures that the study remains focused on providing **decision-relevant insights** for UNHCR while delivering a clear, actionable recommendation for renewable energy deployment in the camp.

4. Assumptions and Methodology

This section describes the approach used to assess the technical and economic feasibility of an off-grid solar photovoltaic (PV) and battery energy storage system for a refugee camp near Damascus, Syria. It outlines the methodology for load estimation, system sizing, and economic evaluation, as well as the key assumptions underpinning the analysis.

4.1 Methodology

4.1.1 Load Estimation

- The total energy demand for the camp was calculated by aggregating loads from **residential prefab units, clinics, schools, street lighting, and water pumping stations**. For each category, the following formula was applied:

Daily Energy (kWh) = Number of Units × Rated Power (kW) × Daily Hours of Operation

- Residential loads include TVs, fans, LED lights, and small refrigerators. Communal loads include clinics (medical equipment and lighting), schools, street lighting, and water pumping stations.
-

4.1.2 System Sizing

- **PV Array Sizing:**

$$\text{PV Capacity (kWp)} = \frac{\text{Total Daily Energy Demand (kWh)}}{\text{Solar Irradiance (kWh/kWp/day)} \times \text{System Efficiency}}$$

- **Battery Storage Sizing:**

$$\text{Battery Capacity (kWh)} = \frac{\text{Daily Energy Demand (kWh)} \times \text{Autonomy Days}}{\text{Depth of Discharge (DoD)} \times \text{Round-Trip Efficiency}}$$

- Autonomy days = 1 day
 - Depth of Discharge (DoD) = 80%
 - Round-trip efficiency = 90%
 - **Inverter Sizing:**
Inverter capacity is sized to match **peak load**, with a 10–15% margin for safety.
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4.1.3 Economic Analysis

The economic assessment compares the **solar PV + battery system** with a **diesel generator alternative**, considering:

- **Capital Expenditures (CapEx):** PV modules, batteries, inverters, balance of system (BOS), and installation.
 - **Operating Expenditures (OpEx):** Annual O&M costs for PV, battery replacements, and diesel generator fuel and maintenance.
 - **Levelized Cost of Energy (LCOE):** Calculated over a 20-year project life.
 - **Payback Period and Total Cost of Ownership (TCO):** For decision-making and comparison with diesel generation.
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4.2 Key Assumptions

Parameter	Value	Notes
Population	About 100,00	5 people per prefab unit
Household energy use	3.03 kWh/day	TV, fans, LED lights, refrigerator
Clinic load	0.25 MWh/day	2 refrigerators, 10 LED lights, medical equipment
Water pumping	75 kW motors × 24 h	Continuous operation assumed
School lighting	500 LED lights × 5 h/day	Duty cycle = 5 h/day
Street lighting	100 LED lights × 10 h/day	Typical street lighting schedule
Solar irradiance	5.5 kWh/m ² /day	Damascus region (Global Solar Atlas)
PV panel cost	\$300/kWp	Module only
Battery cost	\$200/kWh	Li-ion, 10-year lifetime, 80% DoD, 90% efficiency
Diesel fuel cost	\$1.20/L	Regional Syria market
Diesel generator CAPEX	\$500/kW	Approximate
System lifetime	20 years	PV modules 25 yrs, batteries replaced at 10 yrs
Discount rate	6%	Standard for UNHCR projects
Autonomy	1 day	Ensures power supply during low solar periods
System efficiency	85%	Includes inverter and derating losses

4.3 Notes

- All calculations assume **daily energy demand profiles** with constant operating hours per load category.
- Costs are presented in **2025 USD (most current data used)**.
- Diesel generator assumptions are based on **regional market reports** and typical fuel consumption for medium-scale generators (0.26 L/kWh).
- The analysis does **not include grid connection or civil works** beyond PV and battery installation.

5. Technical Design

This section details the design of the **off-grid solar PV and battery energy storage system** for the refugee camp near Damascus. The design is based on the load estimates, assumptions, and methodology presented previously

5.1 Load Summary

The total daily energy demand for the camp was calculated by aggregating all major loads. The results are summarized in the table below:

Load Category	Units	Rated Power (W/unit)	Hours/day	Daily Energy (kWh)
Residential prefab units	20,000	320	6	38,400
Clinic	2	10,500	8	168
Water pumping	3 motors	25,000	24	1,800
School lighting	500	10	5	25
Street lighting	100	150	10	150

TOTAL estimate: 40.5 MWh/day

5.2 PV System Sizing

Using the formula:

$$\text{PV Capacity (kWp)} = \frac{\text{Total Daily Energy (kWh)}}{\text{Solar Irradiance (kWh/kWp/day)} \times \text{System Efficiency}}$$
$$\text{PV Capacity} = \frac{40,543}{5.5 \times 0.85} \approx 8,660 \text{ kWp (8.7 MWp)}$$

- **PV panels:** \$300/kWp
- **Total PV cost:** 8.66 MW × \$300/kWp = ~\$2.6 million

Note: This PV sizing ensures sufficient energy for daily consumption, accounting for system losses and derating.

5.3 Battery Storage Sizing

The battery storage was sized for **1 day of autonomy** with **Depth of Discharge (DoD) = 80%** and **round-trip efficiency = 90%**:

$$\text{Battery Capacity (kWh)} = \frac{40,543 \times 1}{0.8 \times 0.9} \approx 56,200 \text{ kWh (56 MWh)}$$

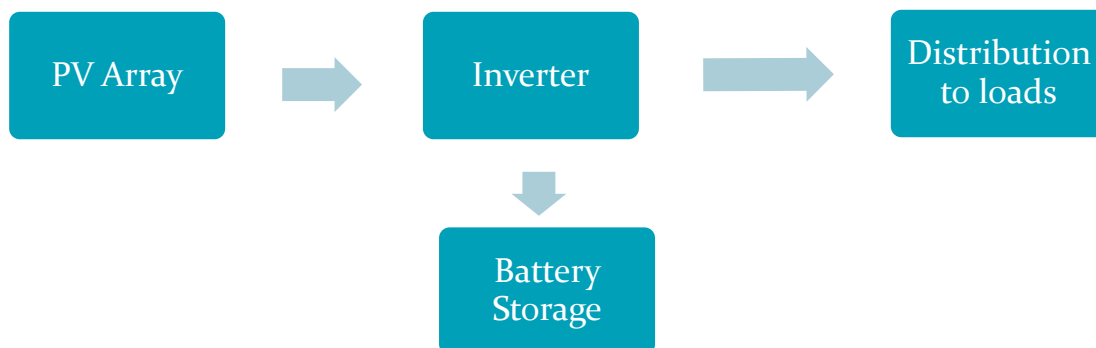
- **Battery type:** Li-ion, 10-year lifetime
- **Battery cost:** 56,200 kWh × \$200/kWh ≈ \$11.2 million

Note: Batteries will need replacement at year 10 for a 20-year project.

5.4 Inverter Sizing

Peak load is estimated at **4 MW**, which includes simultaneous operation of residential, pumping, and clinic loads.

- **Inverter capacity:** 4.5 MW (10–15% margin for safety)
- **Inverter cost:** Assumed \$150/kW → 4,500 kW × \$150 = \$0.675 million



6. Economic Analysis

This section compares the **off-grid solar PV + battery system** with a **diesel generator alternative**, taking into account capital expenditures (CapEx), operating costs (OpEx), fuel costs, and lifetime ownership costs. The analysis helps determine the economic feasibility of adopting a renewable energy solution for the refugee camp.

6.1 Solar PV + System Cost

Component	Capacity	Cost per unit	Total Cost (USD)	Notes
PV array	8.66 MWp	\$300/kWp	\$2,598,000	Includes modules only
Battery Storage	56 MWh	\$200/kWh	\$11,240,000	Li-ion, 1-day autonomy
Inverters	4.5 MW	\$150/kW	\$675,000	Peak load + 10–15% margin
BOS & Installation		15% of total	\$2,960,000	Mounting, wiring, installation

TOTAL estimate: \$17,473,000 (includes: PV, Battery, Inverter and BOS)

- **Operations & Maintenance (O&M):** \$200,000/year (panels, inverters, batteries)
- **Battery Replacement:** Year 10 → ~\$11.2M

6.2 Diesel Generator Alternative Costs

Component	Capacity	Cost per Unit	Total Cost (USD)	Notes
Diesel Generators	4.5 MW	\$500/kW	\$2,250,000	Peak load requirement
Fuel Cost	4,000 kWh/day × 0.26 L/kWh × \$1.20/L		\$456,000/year	Average fuel consumption for medium diesel generators
O&M			\$150,000/year	Routine maintenance

TOTAL estimate: \$13.8M + \$12.12M fuel = \$25.9M

Note: An average cost of fuel per liter is used with most current data

6.3 Levelized Cost of Energy (LCOE)

Solar PV + Battery LCOE: $17.47 + 11.24 + (200,000 \times 20) / 40.5 \text{ MWh/day} \times 365 \times 20$
 $\approx 0.18 \text{ USD/kWh}$

Diesel LCOE: $25.9\text{M} / 40.5 \text{ MWh/day} \times 365 \times 20$
 $\approx 0.25 \text{ USD/kWh}$

6.4 Payback Period

- **Initial CapEx difference: Diesel \$2.25M vs Solar \$17.47M**
- **Annual savings: Diesel fuel + O&M (\$606,000/year) – Solar O&M (\$200,000/year) = \$406,000/year**
- **Payback Period:**

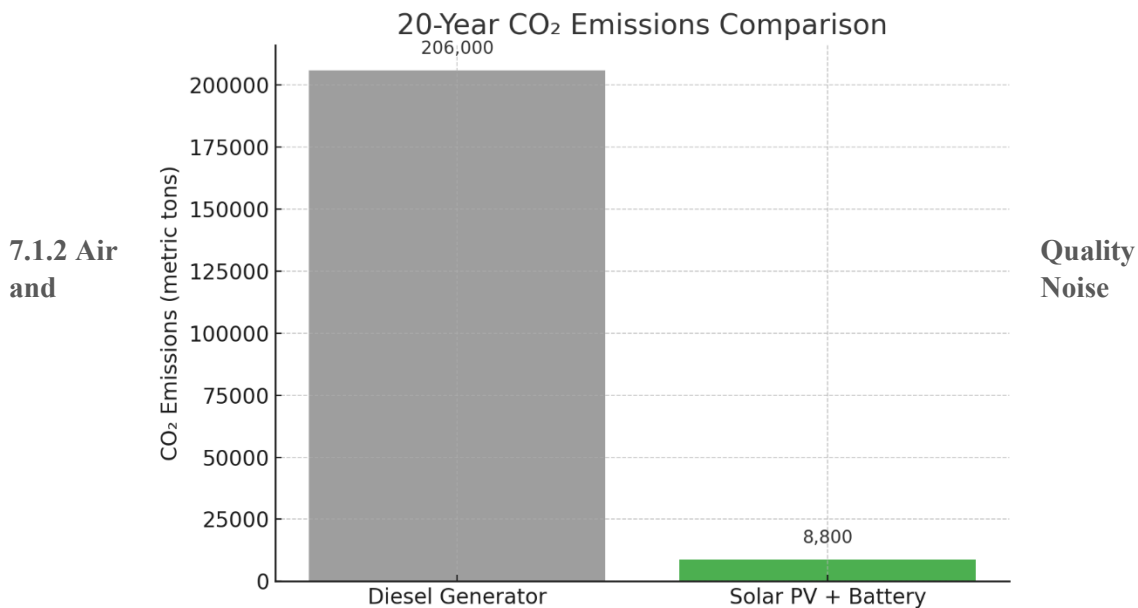
$$\text{Payback} = \frac{\text{Initial Investment Solar} - \text{Diesel}}{\text{Annual Savings}} = \frac{17.47 - 2.25}{0.406} \approx 37 \text{ years}$$

7. Environmental and Social Impacts

7.1.1 Greenhouse Gas Emission Reduction

Diesel power generation emits approximately **2.68 kg CO₂ per liter** of fuel burned. Based on the camp's annual energy demand of roughly **14,782 MWh/year** and a diesel fuel consumption rate of **0.26 L/kWh**, the baseline system would release an estimated **~10,300 metric tons of CO₂ annually**.

In contrast, the solar PV + battery system produces **near-zero emissions during operation** and only **~440 metric tons of CO₂ per year** when considering life-cycle manufacturing impacts. Over a 20-year period, this transition results in an **emissions reduction of approximately 197,000 metric tons of CO₂**.



Improvements

Diesel generators produce significant **particulate matter (PM_{2.5})**, **nitrogen oxides (NO_x)**, and **sulfur oxides (SO_x)**, which negatively affect respiratory health. Replacing diesel with PV and battery systems eliminates local air pollution and substantially reduces **noise emissions**, improving living conditions across the camp.

7.1.3 Reduced Fuel Logistics and Risk

Diesel delivery requires frequent transport of fuel through or around conflict-affected regions, increasing cost, supply insecurity, and safety risks. The solar-battery system minimizes these risks by providing **energy independence**, reducing reliance on fuel convoys and storage infrastructure.

7.2 Social and Community Impacts

1. Health and Well-Being:

Improved air quality and reduced noise enhance physical and mental health within the camp.

2. Employment and Capacity Building:

The construction, operation, and maintenance of the solar facility will create local job opportunities and skill development in renewable-energy technologies.

3. **Education and Stability:**

Reliable electricity supports extended school hours, better lighting for studying, and reliable power for clinics and communications.

4. **Resilience and Security:**

Energy self-sufficiency strengthens the camp's resilience to regional disruptions, ensuring continuity of essential services during crises.

8. Conclusions and Recommendations

8.1 Conclusions

The techno-economic assessment confirms that an **off-grid solar photovoltaic (PV) system combined with battery energy storage** is a technically feasible and economically favorable solution for supplying electricity to the refugee camp near Damascus, Syria.

🔍 **Technical Feasibility:**

The proposed system—approximately **8.7 MWp of solar PV** and **56 MWh of battery storage**—can reliably meet the camp's total daily energy demand of ~40.5 MWh. The design ensures one full day of autonomy and continuous 24-hour power availability for critical facilities such as clinics and water pumps.

🔍 **Economic Performance:**

The solar-battery system achieves a **Levelized Cost of Energy (LCOE) of $\approx$ \$0.18/kWh**, compared to **\$0.25/kWh** for diesel generation. Although the PV-battery system's initial capital cost is higher, it delivers significant long-term savings through reduced fuel and maintenance costs.

🔍 **Environmental Impact:**

Transitioning from diesel generation to solar PV + storage will **reduce CO₂ emissions by approximately 197,000 metric tons over 20 years**, improve air quality, and eliminate fuel-supply risks.

🔍 **Social Benefits:**

Cleaner energy enhances living conditions, supports educational and healthcare services, and creates local employment opportunities in installation and maintenance.

8.2 recommend steps

🔍 **Phased Implementation:**

Begin with a **10 MWp pilot installation** (approximately 35% of total design) to validate performance and operational logistics. Scale up to full deployment in stages over 18 months.

🔍 **Hybrid Backup Strategy:**

Retain a **small modular diesel backup (500–1000 kW)** for emergency use to maintain power during maintenance or extended cloudy periods.

🔍 **Local Capacity Building:**

Partner with Syrian technical institutes or NGOs to train local staff for operation and maintenance, promoting community ownership and job creation.

🔍 **Battery Replacement Planning:**

Budget for **battery renewal in year 10** of operation and negotiate long-term supplier warranties (minimum 10-year performance guarantee).

🔍 **Monitoring and Data Collection:**

Implement a centralized energy-management system to track generation, consumption, and storage performance, ensuring continuous optimization and transparency for donors.

🔍 **Funding and Partnerships:**

Explore collaboration with international development banks, renewable-energy NGOs, and humanitarian agencies to secure financing and reduce upfront investment.

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